Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC

Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Global Markets Strategy

23 October 2026

J.P.Morgan

Treasuries

AI, OATs, and oil, oh my

- Treasury yields have displayed flight-to-quality properties amid stress in OATs and weakness in tech stocks. Looking ahead, we think this week's equity move was more technically driven, valuations once again appear rich, and Treasuries have stalled near resistance, which suggests near-term room for yields to move higher...
- We expect to learn little from next week's FOMC meeting: given the Fed hiked off a 0.2% core PCE at the last meeting and is set to hold off a firmer reading this month, markets could interpret this dovishly, leaving room for a twist steepening
- 10s/30s continues to appear flat given the level of medium-term Fed policy expectations and we recommend holding tactical 10s/30s steepeners
- We examine recent trends in primary demand, finding that end-user auction demand has continued to hold up well in 2026 even amid broad market volatility, with demand at the long-end especially strong...
- ...Demand at auction has been bolstered by increased foreign participation, especially in longer tenors, suggesting the demand is likely primarily foreign private investors. The data suggest foreign demand remains solid, despite broad "de-dollarization" fears...
- ...Auction tails have increased in most tenors, which could reflect a shift to somewhat more price sensitive foreign investors, or may be a natural result of higher intraday volatility. Meanwhile, long-end auction tails have fallen this year; with demand in the sector strong, we do not see cuts to long-end auction sizes as warranted

Market views

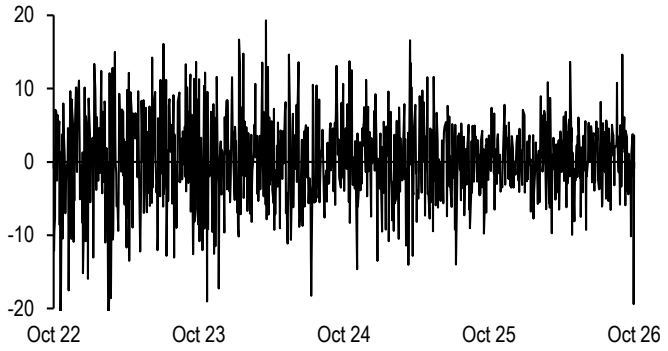
Treasury yields declined 5-12bp this week with the long end leading the way. Substantially all of this move occurred on Wednesday, which marked the largest one-day decline since the regional banking failures of March 2023 (**Figure 1**). Against the backdrop of these moves, money markets have priced in a less hawkish path for Fed policy over the next year: OIS forwards are now pricing in 70bp of hikes by next fall (down from a peak of more than 90bp at the end of September), and a full 25bp hike is no longer priced for the December FOMC meeting (**Figure 2**).

The move was dictated by increased risk aversion, amid signs of stress in European government bond markets and concerns over AI-related equity valuations, and has also been driven by changing sensitivities to ongoing increases in energy prices. First, the OAT/Bund spread widened 35bp this week, to levels seen only briefly in the fall of 2011 amid the worst of the European peripheral crisis. As our colleagues in Europe write, there has been no catalyst for this week's widening, though the escalation of French protests last weekend likely worsened sentiment amid historically weak liquidity. France is screening as cheap in our fiscal framework as Italy did for much of the prior decade, but they expect spread vol to remain elevated and think it would require some combination of an improvement in French fiscal or political dynamics, or signs of ECB signaling on potential intervention to break this cycle (see *France: catalysts to watch for a reversal in spreads*, Aditya Chordia, 10/22/26).

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Figure 1: 10-year yields had their largest daily decline since the regional bank failures of March 2023, amid further OAT/Bund widening and weakness in tech stocks...

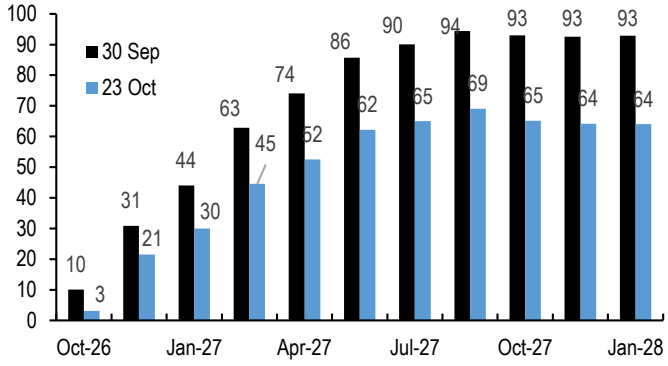
Daily changes in 10-year Treasury yields; bp



Source: J.P. Morgan

Figure 2: ...and OIS forwards have taken more than one full hike out of market pricing over the last three weeks

Expected change in Fed funds effective rate by FOMC meeting implied by OIS forwards, 9/30/26 vs. 10/23/26; bp



Source: J.P. Morgan

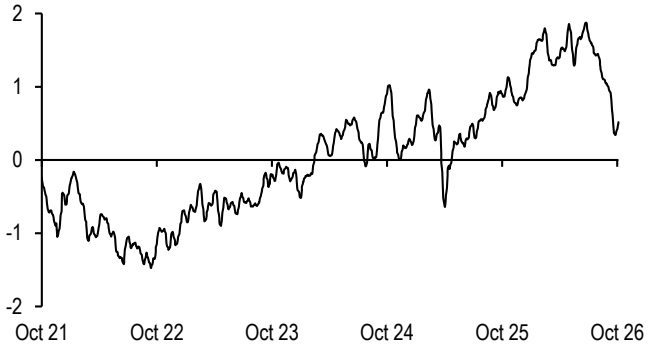
Second, Treasuries displayed their traditional flight-to-safety qualities in this move: while EGB market functioning has worsened substantially, Treasury markets have been more orderly. We have developed a market functioning index, which aggregates measures of Treasury market depth (to measure how much is tradeable), price impact (to measure the impact each trade has on moving markets), and dispersion (to measure how efficiently off-the-run Treasuries are trading to a fitted curve, **Figure 3**). In aggregate, Treasury market functioning has deteriorated as delivered volatility has increased over the last two months, but remains higher than average levels observed over the last three years. This indicates to us that Treasury market liquidity remains more robust than it has been on average over this horizon, which has been a period marked by a structural improvement in Treasury market functioning (see *Ain't nothin' gonna break my stride: Treasury market liquidity continues to improve*, 2/27/26). **In effect, we think the increased stress in EGBs has resulted in incremental demand for Treasuries, which remain the largest and deepest government bond market in the world.**

Finally, we do not want to understate the impact of shifting narratives on domestic risk assets as well. **Figure 4** shows that over the last month, Treasury yields have become increasingly correlated with the performance of the NASDAQ, while displaying a negative correlation with Brent prices. It is difficult to disentangle these two factors, as earlier in the Middle East conflict, Treasury yields began to display a negative sensitivity to oil prices when Brent broke above $110/bbl. Looking ahead, despite a resilient economy, equity markets are likely in the driver's seat for rates valuations. As we have highlighted in recent weeks, our equity team has made the point that rising rates become a headwind for equity multiples, as higher discount rates and higher inflation begin to compress valuations. Depending on the growth regime, this happens between 5 and 6% on 10-year yields (see *Global Equity Strategy: Implications of Higher Rates for Equity Fundamentals, Valuation, and Positioning*, Dubravo Lakos-Bujas, et al., 9/14/26).

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Figure 3: Treasury market functioning has deteriorated amid the rise in volatility, but remains above its medium-term averages in what has been a period of improving liquidity conditions

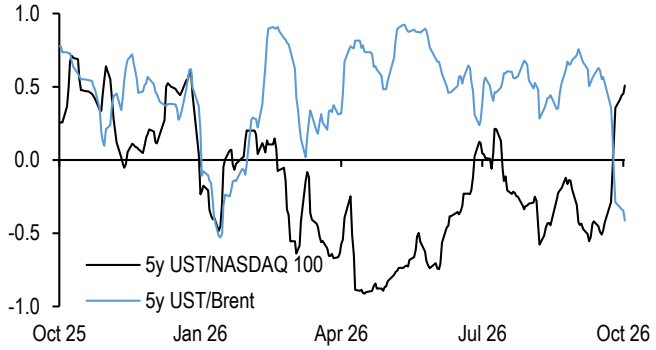
Treasury market functioning index*, 10-day moving average; rolling 3-year z score



* We aggregate Treasury market depth**, price impact†, and Treasury curve RMSE‡, by sector of the curve using 3-year rolling z-scores, take the average across each tenor and then take a duration-weighted aggregate average to produce this series
** Market depth is defined as the sum of the top 3 bids and offers by queue position, averaged between 8:30 and 10:30am; 1-week moving average
† Price impact is the average move in order book mid-price against a $100mn flow in traded notional. See Drivers of price impact and the role of hidden liquidity, 1/13/17 for more details
‡ We average the market depth, price impact, and RMSE z-scores. We multiply the price impact and RMSE z-scores by -1 so positive z-score means better Treasury market functioning
Source: J.P. Morgan

Figure 4: The tech equity sphere has become a dominant driver of Treasury yields in recent weeks

Rolling 1-month correlations of weekly changes in 5-year Treasury yields (bp) versus weekly changes in NASDAQ 100 index (%) and Brent oil futures (%)



Source: J.P. Morgan

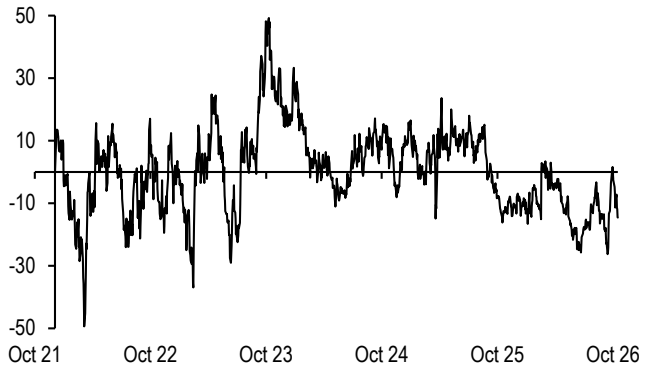
While rates markets participants will likely have more than one eye on the equity market going forward, it's notable that for the time being, Treasuries have stalled near technical resistance levels, and broad equity indices have as well. As long as the S&P 500 maintains above this 7500-7600 range, this should remove one bullish impulse over the near term (see *Global Fixed Income Technical Update*, Jason Hunter, 10/23/26).

Away from equities, the other elephant in the room is the FOMC meeting next Wednesday. We, like the consensus, expect the Fed to stand pat, and markets are pricing in just a 10% probability of a hike next week. We expect two hawkish dissents from Hammack and Logan, with a risk Kashkari dissents as well, but we expect to learn little from the statement or the press conference. Importantly, the Chair will have to explain why the Committee was willing to hike over a 0.2% core PCE reading in August, but not what is likely to be a 0.3% print in September: the rest of the Committee has argued it needs more time to monitor the impact of the first hike, but if Warsh defers on this, it's likely market participants will view this as dovish in nature (see *FOMC Preview*, Michael Feroli, 10/23/26). **Given these factors, we think risks to Treasury yields skew modestly higher over the near term, especially given that 10-year yields are once again flagging 15bp too low in our fair-value framework (Figure 5).**

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Figure 5: Treasuries again appear somewhat expensive after controlling for their fundamental drivers...

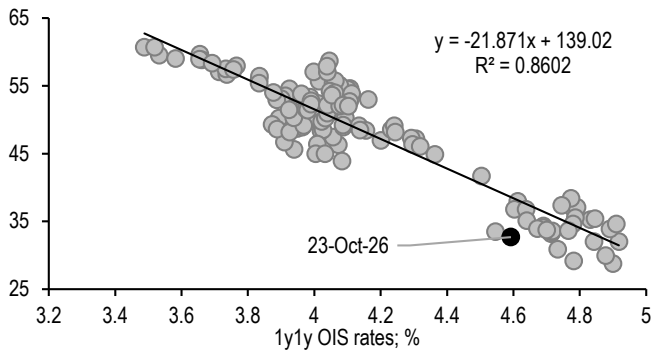
Residual of J.P. Morgan 10-year Treasury fair value model*; bp



*10-year Treasury yields regressed on 1y1y OIS rates (%), 5y5y seasonally-adjusted TIPS breakevens (%), J.P. Morgan U.S. Forecast Revision Index (%), Fed B/S as share of the US economy (%), and a dummy variable for trade policy uncertainty that is 1 between 4/2/25 and 2/26/26. Regression over the last 5 years: R-squared = 9.3%, SE = 13.9bp
Source: J.P. Morgan, Federal Reserve

Figure 6: The long end has lagged the repricing in medium-term Fed expectations: we hold 10s/30s steepeners as a low-beta long exposure with some relative value

10s/30s Treasury curve regressed on 1y1y OIS rates (%), regression over the last 6 months; bp



Source: J.P. Morgan

Against this backdrop, we think the curve could twist steeper next week, and we continue to like long-end steepeners as a low-beta way to position for lower front-end yields with some relative value. The 10s/30s curve is largely driven by medium-term Fed expectations, and has lagged the repricing in Fed expectations that has occurred over the last two weeks; on this basis, 10s/30s appears about 6bp too flat (**Figure 6**). **Accordingly, we continue to recommend holding 10s/30s steepeners** (see *US Treasury Market Daily*, 10/15/26).

Primary demand update

Primary demand has been the topic du jour as of late, especially as Treasury has highlighted medium-term demand trends as a key consideration when evaluating future adjustments to auction sizes. With the September auction allotment data now in hand, we can examine the evolution of primary demand through 3Q26. Despite concerns about demand weakness amid broad geopolitical and market volatility this year, end-user demand has broadly held up quite well. This has been particularly true for longer tenors: average end-user demand at the 10- and 30-year auctions is the highest on record, at 90.6% and 90.1%, respectively (**Figure 7**). While end-user demand in the belly has experienced some pullback on a year-over-year basis from the record highs observed in 2025, demand remains at very strong levels historically.

Primary demand has been largely bolstered by foreign participation, which surged in March alongside the start of the Middle East conflict, and stepped up sharply once again in September as yields repriced to multi-decade highs in a volatile fashion (**Figure 8**). Indeed, the mid-month September auctions saw some of the highest rates of foreign participation on record. The fact that this increase has occurred primarily in the belly to long end says to us that this demand is likely emanating from the foreign private sector, whose holdings tend to be concentrated further out the curve than those of official investors, as official holdings are kept shorter in duration so they can be more easily liquidated in times of financial stress. This follows similar trends in the August TIC data, which showed foreign investors net bought long-term Treasuries in August at their fastest pace in over a year, primarily driven by an uptick in foreign private demand (see *US Treasury Market Daily*, 10/19/26). **Overall,**

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

we find it striking that despite prevalent fears of broad "de-dollarization," the data suggests foreign participation in the Treasury market remains strong. It is telling that in times of stress, global investors continue to reach for Treasuries, as they remain the deepest and most liquid sovereign debt market in the world.

Figure 7: End-user demand has held up well this year, especially in the long end...

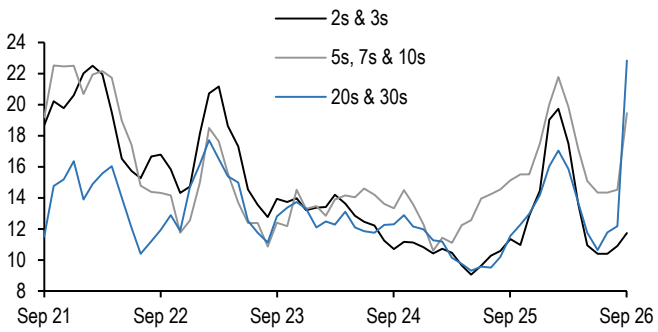
Average annual* share of end-user demand by tenor; %

	Nominals						
Year	2y	3y	5y	7y	10y	20y	30y
2019	69.6	64.4	76.2	78.7	74.5		74.7
2020	66.2	63.7	74.2	77.8	74.5		78.6
2021	71.5	69.9	76.4	78.2	82.8	79.2	81.4
2022	80.2	77.3	82.2	86.7	83.2	88.2	86.9
2023	83.3	82.7	86.9	87.0	85.6	89.8	86.6
2024	86.6	83.6	85.5	88.0	85.4	86.3	85.2
2025	88.7	86.4	89.3	89.6	89.0	87.6	86.8
2026	*87.9*	*86.6*	*87.0*	*88.2*	*90.6*	*88.4*	*90.1*
Diff*	***-0.8***	***0.2***	***-2.3***	***-1.4***	***1.5***	***0.8***	***3.3***

* 2026 data incorporates auctions through mid-month October supply process
Source: US Treasury

Figure 8: ...bolstered by a sharp increase in foreign participation this year

Foreign share of demand by sector; 3-month moving average



Source: US Treasury

Interestingly, strong demand has not translated to smaller auction tails, as average auction tails have grown in 2026 across nearly all tenors (**Figure 9**). We see a couple of possible explanations for this. **First**, foreign private investors, which now make up a larger share of primary demand, may display greater price sensitivity at auction than their less price-sensitive official sector counterparts. This is a similar dynamic to one that we have seen play out across the wider Treasury market over the last several years.

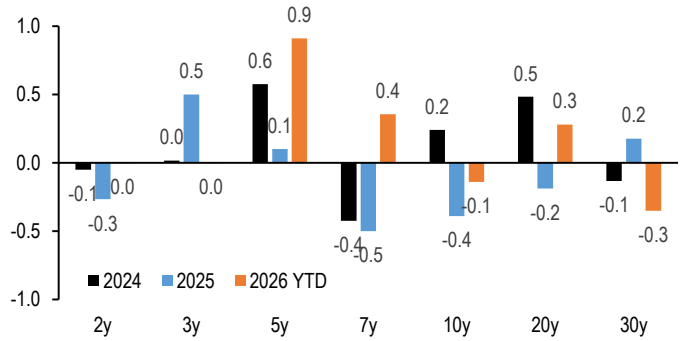
Second, increased rates volatility, and especially intraday delivered vol, has naturally led to larger tails. Auction tails largely increased in March as the conflict drove broad market vol, and spiked again in September as rates repriced (**Figure 10**). Once again, the long end is the standout exception to the trend, which is especially notable in the context of the general narrative that Treasury could adjust long-end issuance, potentially due to demand trends in the sector — we already view such a decision as imprudent in the context of widening financing gaps over the coming years, and the fact that long-end auctions have held up so well similarly does not suggest such a move would be warranted.

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

J.P.Morgan

Figure 9: Despite strong end-user demand, average auction tails have grown this year, with the exception of the long end

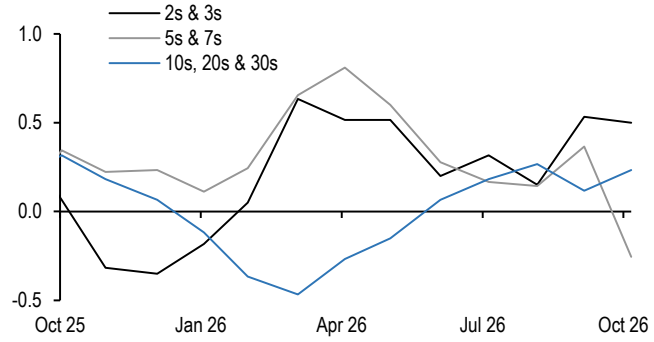
Average auction tails by tenor*; bp



*2026 data incorporates auctions through mid-month October supply process
Source: US Treasury

Figure 10: Auction tails have tended to increase in line with periods of broader market volatility

Average auction tails by sector*, 3-month moving average; bp



* October 2026 averages include only the realized 3-, 10-, 20-, and 30-year auctions
Source: US Treasury

Trade recommendations

- **Maintain 10s/30s steepeners**

- Stay long 100% risk, or $64.8mn notional of T 4.625% Aug-36s
- Stay short 100% risk, or $35.9mn notional of T 5.125% Aug-56s
- (*US Treasury Market Daily*, 10/15/26: P/L since inception: -1.4bp)

- **Maintain 10s/20s/30s belly-cheapening butterflies**

-Stay long 50% risk, or $32.1mn notional of T 4.625% Aug-36s
-Stay short 100% risk, or $41.2mn notional of T 5.125% Aug-46s
-Stay long 50% risk, or $16.9mn notional of T 5.125% Aug-56s
- (*US Treasury Market Daily*, 9/4/26: P/L since inception: -2.9bp)

- **Maintain 2.375% May-51 / 5.0% May-56s curve flatteners**

- Stay short 100% risk, or $25.1mn notional of T 2.375% May-51s
- Stay long 100% risk, or $16.6mn notional of T 5% May-56s
- (*US Fixed Income Markets Weekly*, 5/15/26: P/L since inception: -1.1bp)

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC
Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Figure 11: Closed trades in the last 12 months

P/L reported in bp of yield unless otherwise indicated

TRADE	ENTRY	EXIT	P/L
Duration			
50:50 weighted 2s/5s/10s belly-cheapening butterflies	11/25/25	12/10/25	4.9
2-year duration shorts	02/12/25	03/05/26	13.9
2-year duration longs	03/20/26	04/02/26	9.4
50:50 weighted 2s/5s/10s belly-cheapening butterflies	03/06/26	05/15/26	3.7
2-year duration longs	05/15/26	06/04/26	3.8
Short 10-year Treasuries versus 10-year Bunds	06/05/26	09/25/26	13.8
Curve			
5s/20s steepeners	07/25/25	11/24/25	0.6
2s/5s flatteners	12/16/25	01/05/26	-2.6
10s/30s flatteners	01/09/26	01/20/26	1.9
5s/20s steepeners	01/28/26	02/06/26	6.2
5s/20s steepeners	02/20/26	03/11/26	0.7
10s/30s flatteners	06/12/26	07/13/26	-0.4
5s/30s flatteners	07/24/26	07/29/26	-5.8
2s/10s steepeners	07/31/26	08/28/26	-7.6
Relative value			
67:37 4.00 Jun-32s/4.25 Aug-35s/4.75 Aug-55s belly-cheapening butterflies	10/03/25	12/19/25	-3.4
50:50 weighted 5s/10s/30s belly-cheapening butterflies	10/31/25	06/12/26	-5.8
Term premium			
64:11 weighted 3s/10s/30s belly-cheapening butterfly	05/02/25	11/24/25	-18.3
Number of positive trades			10
Number of negative trades			7
Hit rate			59%
Aggregate P/L			15.0

Source: J.P. Morgan

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC

Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Digital assets, which may also be referred to as virtual or crypto assets or crypto currencies (e.g. bitcoin), discussed in this research report are subject to a rapidly changing regulatory landscape. The application of relevant laws in respect of such assets may be unsettled, and the legality of any digital asset investment may be unclear. Prior to making any investment decision regarding any digital asset, investors should seek their own legal advice and be aware of any legal considerations regarding the purchase, holding or selling of such an investment. For relevant regulatory advisories on digital assets, including bitcoin and ethereum, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures

Jay Barry [AC] (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Harry Downie (1-212) 270-9500
harry.j.downie@jpmorgan.com
J.P. Morgan Securities LLC

Amanda Berke (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the

Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - 68724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Nigeria: J.P. Morgan Chase Representative Office (Nigeria) Limited is regulated by the Central Bank of Nigeria.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 070/08/2026 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to

engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumphini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is

subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Any mentions of third-party trademarks, brand names, products, and services are for referential purposes only and any mention thereof is not meant to imply any sponsorship, endorsement, or affiliation.

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised October 17, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. Without prior written consent from J.P. Morgan, it is prohibited to (A) use or share any material, commentary, information or other data received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any artificial intelligence ("AI") systems or models, or (B) use automated means, including robots, crawlers, web-scrapers or other automated tools or scripts, to access, collect, extract or harvest any J.P. Morgan Data, including scraping of user data, content, metadata, pricing information or other information.

Completed 23 Oct 2026 04:51 PM EDT

Disseminated 23 Oct 2026 04:52 PM EDT